Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/05/2018
PI Name:	Rosenblatt, Charles S. Ph.D.		
Project Title:	Microgravity Studies of Nanoparticles and Topological Defects in Liquid Crystal Thin Films		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/02/2016	**End Date:**	12/01/2021
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Tin, Padetha	**Contact Phone:**	216-433-8164
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Flight Program:	ISS		
Flight Assignment:	Liquid Crystal Facility		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lacaze, Emmanuelle Ph.D. (Université Pierre et Marie Curie) Hegman, Torsten Ph.D. (Kent State University) Taylor, Philip Ph.D. (Case Western Reserve University) Lemieux, Robert Ph.D. (COLLABORATOR / University of Waterloo)		
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Task Description:	Nanoparticle / liquid crystal composites have created new avenues for scientific discovery and technological development. For example, recent studies have shown that mixtures of liquid crystals and appropriate semiconducting nanoparticles can be exploited for potential high efficiency and easily manipulatable photovoltaic devices. But the development of this and other technologies requires a fundamental understanding of how the nanoparticles incorporate and move within the liquid crystal, how they aggregate in and around topological defects, and how the nanoparticles affect the defects themselves. We have shown that fluorescent carbon dots (FCDs) may be dispersed in liquid crystals and used as a probe of defects, demonstrating that they accumulate in regions of high liquid crystal curvature so as to relieve the local strain. As a result their emitted light can serve as a quantitative indicator for the interaction and evolution of topological defects. Interpretation of these initial results, however, is compromised by the presence of substrates, which is required in an Earth-bound environment. Here we will perform experiments in the absence of a container by utilizing the Observation and Analysis of Smectic Islands In Space (OASIS) module aboard the International Space Station (ISS). By studying thin flate liquid crystal films doped with FCDs in a microgravity environment, we will obviate the need for bounding substrates, thereby rendering a much simpler system for interpretation. In particular, we will use OASIS' cameras to study the spatial distribution and transport of the FCDs in and around topological defects in the smectic-C phase of the liquid crystal, especially close to the smectic-A phase transition temperature. The work will involve both experimental and theoretical components. The results will provide a clear picture of how nanoparticles distribute themselves over time, as well as how defects in the liquid crystal interact.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Liquid crystal (LC) textures and topological defects have been of interest for well over a century. Defects in two-dimensions generally are categorized in terms of the "defect strength"-- "s". Studies of defects can be used to obtain values of LC anchoring strength coefficients and elastic constants, albeit with limited accuracy. Defects are central in establishing the director profile in geometrically-confined liquid crystals, and serve as an important signature when determining the symmetry of phases, for example, distinguishing between a biaxial nematic phase and a Smectic-C (Sm-C) phase. Defect dynamics provide another important field of study, as defect motion is extremely sensitive to boundary effects and provides information about surfaces and impurities. Perhaps most important use of topological defects is that they can serve as traps for micro- and nanoparticles (NPs), as the presence of the particles reduces or eliminates the divergent energy associated with the defect core. In other words, nanoparticles (NPs) migrate within the LC in the direction of increasing stress associated with the liquid crystal's orientational order, as the presence of the particles provides relief from the large stresses that would be present at LC singularities. The incorporation of NPs into liquid crystals and their subsequent behavior in and around defects finds applications in areas such as energy harvesting, photonic structures, mirrorless lasing, micro-patterned LC devices, and chiral metamaterials for photonics and communications applications. They also may have applications in pharmaceuticals, as an alternative means of drug delivery.
Task Progress:	The second year of the project has focused on the creation of free-standing smectic-C films and deposition of the fluorescent carbon dots (FCDs). We accomplished the following: 1. We switched the shape of the liquid crystal (LC) film from a hollow bubble to a flat film of thickness approximately 100 nm (30 layers of smectic LC). This was done because of the inability to track defect motion on the bubble surface, whereas a flat film is amenable to camera tracking software. 2. We found that using an initial mixture of LC and FCDs to create the film resulted in aggregation of the FCDs. Instead, we switched to a system whereby the smectic LC film is created and then an inkjet printer head is used to deposit a mixture of FCDs and solvent onto the film. The solvent then evaporates, leaving the FCDs to incorporate into the film. We also investigated the efficacy of the inkjet deposition angle onto the film, finding little difference when varying the angle from 90 degrees (perpendicular to the film) down to 45 degrees. We have designed and have 3D-printed a holder for the inkjet nozzle that will provide better protection against damage, and facilitate angular and positional control. 3. We investigated various solvents for the FCD / solvent mixture, and found that methanol is an effective and safe choice. The FCDs do not aggregate in the methanol, and the FCDs prefer the LC to the methanol after deposition onto the LC film. 4. We investigated the optical properties of several LCs and molecules attached to the FCDs to make sure that we do not see any unwanted artifacts in our fluorescent signal. 5. We examined how best to eliminate unwanted smectic "islands" from the film. We found that our original plan of using air blowers to push islands to the periphery would not be adequate. Instead, we found that the new film holder design by Hiroshi Yokoyama (Kent State U.) could create stable island-free films. We have adopted the new design. 6. We have been looking into the issue of cosmic ray artifacts on the LC and the imaging system. Since the number of high energy electrons is small, we don't expect them to have a deleterious effect on the LC. For the few spurious cosmic ray events on the CCD array, they tend to affect only 1 or 2 random pixels for short bursts (~ nanoseconds), so artifacts in the video will be easy to eliminate using post-data-collection processing. There is a well developed literature on how to handle this. 7. One problem has been the lack of robustness of the Microfab inkjet printer heads. We have found that they are damaged easily, and have been working on methods to avoid having the heads become clogged with the solvent NP/mixture.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Articles in Peer-reviewed Journals	Murray BS, Kralj K, Rosenblatt C. "Decomposition vs. escape of topological defects in a nematic liquid crystal." Soft Matter. 2017 Nov 22;13(45): 8442–50. https://doi.org/10.1039/c7sm01954h ; PubMed PMID: 29083004; PubMed Central PMCID: PMC5699935 , Nov-2017

Articles in Peer-reviewed Journals	Kurioz P, Kralj M, Murray BS, Rosenblatt C, Kralj S. "Nematic topological defects positionally controlled by geometry and external fields." Beilstein J Nanotechnol. 2018 Jan 10;9:109-18. eCollection 2018. https://doi.org/10.3762/bjnano.9.13 ; PubMed PMID: 29441256; PubMed Central PMCID: PMC5789434 , Jan-2018
Dissertations and Theses	Murray BS. "Anchoring-induced topological defects in nematic liquid crystals: core relaxation mechanisms and electro-optics." Dissertation, Case Western Reserve University, Cleveland, Ohio, June, 2018. , Jun-2018